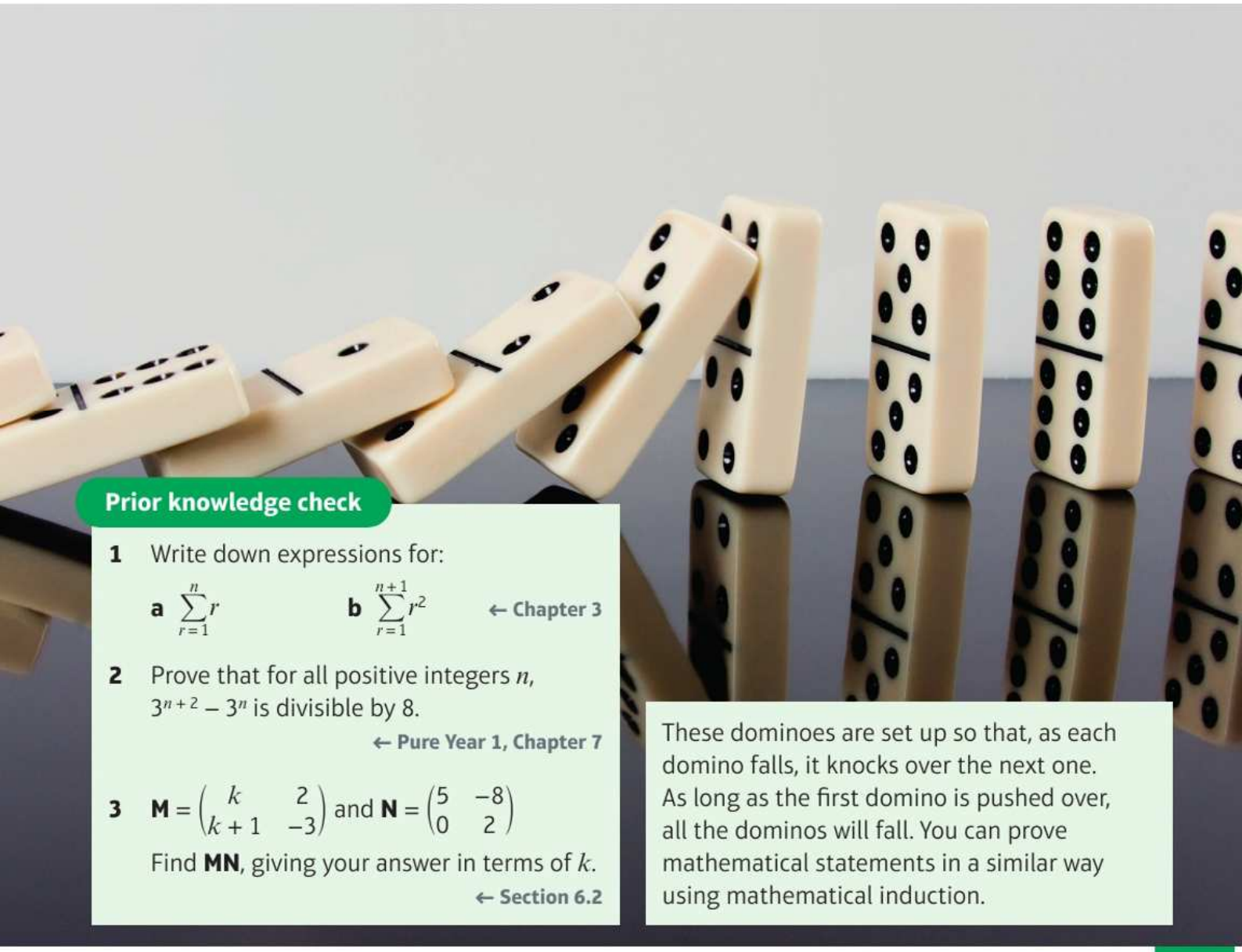
Proof by induction

8

Objectives

After completing this chapter you should be able to:

- Understand the principle of proof by mathematical induction and prove results about sums of series → pages 156–159
- Prove results about divisibility using induction → pages 160–162
- Prove results about matrices using induction → pages 162–164



Prior knowledge check

1 Write down expressions for:

a $\sum_{r=1}^{n} r$ **b** $\sum_{r=1}^{n+1} r^2$ ← Chapter 3

2 Prove that for all positive integers n, $3^{n+2} - 3^n$ is divisible by 8.

← Pure Year 1, Chapter 7

3 $\mathbf{M} = \begin{pmatrix} k & 2 \\ k+1 & -3 \end{pmatrix}$ and $\mathbf{N} = \begin{pmatrix} 5 & -8 \\ 0 & 2 \end{pmatrix}$

Find **MN**, giving your answer in terms of k.

← Section 6.2

These dominoes are set up so that, as each domino falls, it knocks over the next one. As long as the first domino is pushed over, all the dominos will fall. You can prove mathematical statements in a similar way using mathematical induction.

8.1 Proof by mathematical induction

- **You can use proof by induction to prove that a general statement is true for all positive integers.**
- **Proof by mathematical induction usually consists of the following four steps:**

Step 1: Basis: **Prove the general statement is true for $n = 1$.**

Step 2: Assumption: **Assume the general statement is true for $n = k$.**

Step 3: Inductive: **Show that the general statement is then true for $n = k + 1$.**

Step 4: Conclusion: **The general statement is then true for all positive integers, n.**

Watch out You need to carry out **both** the basis step **and** the inductive step in order to complete the proof: carrying out just one of these is not sufficient to prove the general statement.

This method of proof is often useful for proving results about sums of series.

Links You can prove the general results for $\sum_{r=1}^{n} r$, $\sum_{r=1}^{n} r^2$ and $\sum_{r=1}^{n} r^3$ by induction. ← Chapter 3

Example 1

Prove by induction that for all positive integers n, $\sum_{r=1}^{n}(2r - 1) = n^2$.

$n = 1$: LHS $= \sum_{r=1}^{1}(2r - 1) = 2(1) - 1 = 1$

RHS $= 1^2 = 1$

As LHS = RHS, the summation formula is true for $n = 1$.

Assume that the summation formula is true for $n = k$:

$\sum_{r=1}^{k}(2r - 1) = k^2$

With $n = k + 1$ terms the summation formula becomes:

$$\sum_{r=1}^{k+1}(2r - 1) = \sum_{r=1}^{k}(2r - 1) + 2(k + 1) - 1$$
$$= k^2 + (2(k + 1) - 1)$$
$$= k^2 + (2k + 2 - 1)$$
$$= k^2 + 2k + 1$$
$$= (k + 1)^2$$

Therefore, the summation formula is true when $n = k + 1$.

If the summation formula is true for $n = k$ then it is shown to be true for $n = k + 1$. As the result is true for $n = 1$, it is now also true for all $n \in \mathbb{Z}^+$ by mathematical induction.

1. Basis step
Substitute $n = 1$ into both the LHS and RHS of the formula to check if the formula works for $n = 1$.

2. Assumption step
In this step you assume that the general statement given is true for $n = k$.

3. Inductive step
Sum to k terms plus the $(k + 1)$th term.

This is the $(k + 1)$th term.

Sum of first k terms is k^2 by assumption.

This is the same expression as n^2 with n replaced by $k + 1$.

4. Conclusion step
Result is true for $n = 1$ and steps **2** and **3** imply result is then true for $n = 2$. Continuing to apply steps **2** and **3** implies result is true for $n = 3, 4, 5, \ldots$ etc.

Notation $\mathbb{Z}^+$ is the set of **positive integers**, 1, 2, 3, …. It is equivalent to $\mathbb{N}$, the set of natural numbers.

Example 2

Prove by induction that for all positive integers n, $\sum_{r=1}^{n} r^2 = \frac{1}{6}n(n+1)(2n+1)$.

$n = 1$: LHS $= \sum_{r=1}^{1} r^2 = 1^2 = 1$

RHS $= \frac{1}{6}(1)(2)(3) = \frac{6}{6} = 1$

As LHS = RHS, the summation formula is true for $n = 1$.

Assume that the summation formula is true for $n = k$:

$$\sum_{r=1}^{k} r^2 = \frac{1}{6}k(k+1)(2k+1)$$

With $n = k + 1$ terms the summation formula becomes:

$$\begin{aligned}\sum_{r=1}^{k+1} r^2 &= \sum_{r=1}^{k} r^2 + (k+1)^2 \\ &= \frac{1}{6}k(k+1)(2k+1) + (k+1)^2 \\ &= \frac{1}{6}(k+1)(k(2k+1) + 6(k+1)) \\ &= \frac{1}{6}(k+1)(2k^2 + k + 6k + 6) \\ &= \frac{1}{6}(k+1)(2k^2 + 7k + 6) \\ &= \frac{1}{6}(k+1)(k+2)(2k+3) \\ &= \frac{1}{6}(k+1)((k+1)+1)(2(k+1)+1)\end{aligned}$$

Therefore, the summation formula is true when $n = k + 1$.

If the summation formula is true for $n = k$, then it is shown to be true for $n = k + 1$. As the result is true for $n = 1$, it is therefore true for all $n \in \mathbb{Z}^+$ by mathematical induction.

1. Basis step
Substitute $n = 1$ into both the LHS and RHS of the formula to check if the formula works for $n = 1$.

2. Assumption step
In this step you assume that the result given in the question is true for $n = k$.

3. Inductive step

Sum to k terms plus the $(k + 1)$th term.

Rearrange to get same expression as $\frac{1}{6}n(n+1)(2n+1)$ with n replaced by $k + 1$.

4. Conclusion step
Result is true for $n = 1$ and steps **2** and **3** imply result is then true for $n = 2$. Continuing to apply steps **2** and **3** implies result is true for $n = 3, 4, 5, \ldots$ etc.

Example 3

Prove by induction that for all positive integers n, $\sum_{r=1}^{n} r2^r = 2(1 + (n-1)2^n)$.

$n = 1$: LHS $= \sum_{r=1}^{1} r\,2^r = 1(2)^1 = 2$

RHS $= 2(1 + (1-1)2^1) = 2(1) = 2$

As LHS = RHS, the summation formula is true for $n = 1$.

Assume that the summation formula is true for $n = k$:

$$\sum_{r=1}^{k} r\,2^r = 2(1 + (k-1)2^k).$$

1. Basis step

2. Assumption step

With $n = k + 1$ terms the summation formula becomes:

$$\sum_{r=1}^{k+1} r2^r = \sum_{r=1}^{k} r2^r + (k+1)2^{k+1}$$
$$= 2(1 + (k-1)2^k) + (k+1)2^{k+1}$$
$$= 2 + 2(k-1)2^k + (k+1)2^{k+1}$$
$$= 2 + (k-1)2^{k+1} + (k+1)2^{k+1}$$
$$= 2 + (k - 1 + k + 1)2^{k+1}$$
$$= 2 + 2k2^{k+1}$$
$$= 2(1 + k2^{k+1})$$
$$= 2(1 + ((k+1) - 1)2^{k+1})$$

Therefore, the summation formula is true when $n = k + 1$.

If the summation formula is true for $n = k$, then it is shown to be true for $n = k + 1$. As the result is true for $n = 1$, it is now also true for all $n \in \mathbb{Z}^+$ by mathematical induction.

3. Inductive step

$2^1 \times 2^k = 2^{k+1}$

This is the same expression as $2(1 + (n-1)2^n)$ with n replaced by $k + 1$.

4. Conclusion step

Watch out Induction can prove that a given statement is true for all $n \in \mathbb{Z}^+$, but it does not help you derive statements.

Exercise 8A

E/P **1** Prove by induction that for any positive integer n, $\sum_{r=1}^{n} r = \frac{1}{2}n(n+1)$. **(5 marks)**

E/P **2** Prove by induction that for any positive integer n, $\sum_{r=1}^{n} r^3 = \frac{1}{4}n^2(n+1)^2$. **(5 marks)**

E/P **3 a** Prove by induction that for any positive integer n:

$$\sum_{r=1}^{n} r(r-1) = \frac{1}{3}n(n+1)(n-1)$$ **(6 marks)**

b Hence deduce an expression, in terms of n, for $\sum_{r=1}^{2n+1} r(r-1)$. **(3 marks)**

E/P **4 a** Prove by induction that, for any positive integer n:

$$\sum_{r=1}^{n} r(3r-1) = n^2(n+1)$$ **(6 marks)**

b Hence use the standard result for $\sum_{r=1}^{n} r^3$ to find a value of n such that $\sum_{r=1}^{n} r^3 = 4\sum_{r=1}^{n} r(3r-1)$. **(5 marks)**

P **5** Prove by induction that for any positive integer n,

a $\sum_{r=1}^{n} \left(\frac{1}{2}\right)^r = 1 - \frac{1}{2^n}$ **b** $\sum_{r=1}^{n} r(r!) = (n+1)! - 1$ **c** $\sum_{r=1}^{n} \frac{4}{r(r+2)} = \frac{n(3n+5)}{(n+1)(n+2)}$

E/P **6** The box below shows a student's attempts to prove $\left(\sum_{r=1}^{n} r\right)^2 = \sum_{r=1}^{n} r^2$ using induction.

Let $n = 1$. Then LHS $= \left(\sum_{r=1}^{1} r^2\right) = (1)^2 = 1$, and RHS $= \sum_{r=1}^{1} r^2 = 1^2 = 1$, so that LHS = RHS (Basis step).

Now we assume the statement is true for $n = k$:

$$\left(\sum_{r=1}^{k} r\right)^2 = \sum_{r=1}^{k} r^2$$

and so for $n = k + 1$ the statement is

$$\left(\sum_{r=1}^{k+1} r\right)^2 = \sum_{r=1}^{k+1} r^2$$

Hence, by the principle of mathematical induction, the statement is true for all $n \in \mathbb{Z}^+$.

a Identify the error made in the proof. **(2 marks)**

b Give a counter-example to show that the original statement is not true. **(1 mark)**

E/P **7** A student claims that $\sum_{r=1}^{n} r = \frac{1}{2}(n^2 + n + 1)$, and produces the following proof.

Assume that the statement is true for $n = k$:

$$\sum_{r=1}^{k} r = \frac{1}{2}(k^2 + k + 1)$$

When $n = k + 1$:

$$\begin{aligned}\sum_{r=1}^{k+1} r &= \sum_{r=1}^{k} r + (k + 1)\\ &= \frac{1}{2}(k^2 + k + 1) + (k + 1)\\ &= \frac{1}{2}(k^2 + k + 1 + 2(k + 1))\\ &= \frac{1}{2}((k^2 + 2k + 1) + (k + 1) + 1)\\ &= \frac{1}{2}((k + 1)^2 + (k + 1) + 1)\end{aligned}$$

This is the original formula but with $n = k + 1$. Hence, by the principle of mathematical induction, the statement is true for all $n \in \mathbb{Z}^+$.

a Identify the error made in the proof. **(2 marks)**

b Give a counter-example to show that the original statement is not true. **(1 mark)**

Challenge

Prove by induction that for all positive integers n, $\sum_{r=1}^{n} (-1)^r r^2 = \frac{1}{2}(-1)^n n(n + 1)$

Hint $\sum_{r=1}^{n} (-1)^r r^2 = -1^2 + 2^2 - 3^2 + 4^2 - 5^2 + \ldots$

8.2 Proving divisibility results

You can use proof by induction to prove that a given expression is divisible by a certain integer.

Example 4

Prove by induction that for all positive integers n, $3^{2n} + 11$ is divisible by 4.

Let $f(n) = 3^{2n} + 11$, where $n \in \mathbb{Z}^+$.

1. Basis step

$f(1) = 3^{2(1)} + 11 = 9 + 11 = 20 = 4(5)$, which is divisible by 4.

$f(n)$ is divisible by 4 when $n = 1$.

2. Assumption step

Assume true for $n = k$, so that

$f(k) = 3^{2k} + 11$ is divisible by 4.

3. Inductive step

$$\begin{aligned} f(k+1) &= 3^{2(k+1)} + 11 \\ &= 3^{2k} \times 3^2 + 11 \\ &= 9(3^{2k}) + 11 \end{aligned}$$

$$\begin{aligned} f(k+1) - f(k) &= (9(3^{2k}) + 11) - (3^{2k} + 11) \\ &= 8(3^{2k}) \\ &= 4(2(3^{2k})) \end{aligned}$$

$f(k+1) = f(k) + 4(2(3^{2k}))$

As both $f(k)$ and $4(2(3^{2k}))$ are divisible by 4 then the sum of these two must also be divisible by 4.

4. Conclusion step

Therefore $f(n)$ is divisible by 4 when $n = k + 1$.

If $f(n)$ is divisible by 4 when $n = k$, then it has been shown that $f(n)$ is also divisible by 4 when $n = k + 1$. As $f(n)$ is divisible by 4 when $n = 1$, $f(n)$ is also divisible by 4 for all $n \in \mathbb{Z}^+$ by mathematical induction.

Problem-solving

When proving that an expression $f(n)$ is divisible by r, you can complete the induction step by showing that $f(k + 1) - f(k)$ is divisible by r.

Example 5

Prove by induction that for all positive integers n, $n^3 - 7n + 9$ is divisible by 3.

Let $f(n) = n^3 - 7n + 9$, where $n \in \mathbb{Z}^+$.

1. Basis step

$f(1) = 1 - 7 + 9 = 3$, which is divisible by 3.

$f(n)$ is divisible by 3 when $n = 1$.

2. Assumption step

Assume true for $n = k$, so that

$f(k) = k^3 - 7k + 9$ is divisible by 3.

3. Inductive step

$$\begin{aligned} f(k+1) &= (k+1)^3 - 7(k+1) + 9 \\ &= k^3 + 3k^2 + 3k + 1 - 7(k+1) + 9 \\ &= k^3 + 3k^2 + 3k + 1 - 7k - 7 + 9 \\ &= k^3 + 3k^2 - 4k + 3 \end{aligned}$$

Use the binomial theorem or multiply out three brackets.

$f(k+1) - f(k) = (k^3 + 3k^2 - 4k + 3) - (k^3 - 7k + 9)$
$= 3k^2 + 3k - 6$
$= 3(k^2 + k - 2)$
$f(k+1) = f(k) + 3(k^2 + k - 2)$

As both $f(k)$ and $3(k^2 + k - 2)$ are divisible by 3 then their sum must also be divisible by 3.

Therefore $f(n)$ is divisible by 3 when $n = k + 1$.
If $f(n)$ is divisible by 3 when $n = k$, then it has been shown that $f(n)$ is also divisible by 3 when $n = k + 1$. As $f(n)$ is divisible by 3 when $n = 1$, $f(n)$ is also divisible by 3 for all $n \in \mathbb{Z}^+$ by mathematical induction.

4. Conclusion step

Example 6

Prove by induction that for all positive integers n, $11^{n+1} + 12^{2n-1}$ is divisible by 133.

Let $f(n) = 11^{n+1} + 12^{2n-1}$, where $n \in \mathbb{Z}^+$.

1. Basis step

$f(1) = 11^2 + 12 = 133$, which is divisible by 133.
$f(n)$ is divisible by 133 when $n = 1$.

2. Assumption step

Assume true for $n = k$, so that
$f(k) = 11^{k+1} + 12^{2k-1}$ is divisible by 133.

3. Inductive step

$f(k+1) = 11^{k+1+1} + 12^{2(k+1)-1}$
$= 11^{k+1}(11)^1 + 12^{2k-1}(12)^2$
$= 11(11^{k+1}) + 144(12^{2k-1})$
$f(k+1) - f(k) = (11(11^{k+1}) + 144(12^{2k-1})) - (11^{k+1} + 12^{2k-1})$
$= 10(11^{k+1}) + 143(12^{2k-1})$
$= 10(11^{k+1}) + 10(12^{2k-1}) + 133(12^{2k-1})$
$= 10(11^{k+1} + 12^{2k-1}) + 133(12^{2k-1})$
$f(k+1) = f(k) + 10(11^{k+1} + 12^{2k-1}) + 133(12^{2k-1})$
$= f(k) + 10f(k) + 133(12^{2k-1})$
$= 11f(k) + 133(12^{2k-1})$

$12^{2(k+1)-1} = 12^{2k+2-1}$
$= 12^{2k-1+2}$
$= 12^{2k-1}(12)^2$

Problem-solving

Always keep an eye on what you are trying to prove. You need to show that this expression is divisible by 133, so write $143(12^{2k-1})$ as $10(12^{2k-1}) + 133(12^{2k-1})$.

As both $11f(k)$ and $133(12^{2k-1})$ are divisible by 133 then their sum must also be divisible by 133.

Therefore $f(n)$ is divisible by 133 when $n = k + 1$.
If $f(n)$ is divisible by 133 when $n = k$, then it has been shown that $f(n)$ is also divisible by 133 when $n = k + 1$. As $f(n)$ is divisible by 133 when $n = 1$, $f(n)$ is also divisible by 133 for all $n \in \mathbb{Z}^+$ by mathematical induction.

4. Conclusion step

Exercise 8B

(P) **1** Prove by induction that for all positive integers n:

a $8^n - 1$ is divisible by 7

b $3^{2n} - 1$ is divisible by 8

c $5^n + 9^n + 2$ is divisible by 4

d $2^{4n} - 1$ is divisible by 15

e $3^{2n-1} + 1$ is divisible by 4

f $n^3 + 6n^2 + 8n$ is divisible by 3

g $n^3 + 5n$ is divisible by 6

h $2^n(3^{2n}) - 1$ is divisible by 17

(E/P) **2** $f(n) = 13^n - 6^n$

a Show that $f(k + 1) = 6f(k) + 7(13^k)$. **(3 marks)**

b Hence, or otherwise, prove by induction that for all positive integers n, $f(n)$ is divisible by 7. **(4 marks)**

(E/P) **3** $g(n) = 5^{2n} - 6n + 8$

a Show that $g(k + 1) = 25g(k) + 9(16k - 22)$. **(3 marks)**

b Hence, or otherwise, prove by induction that for all positive integers n, $g(n)$ is divisible by 9. **(4 marks)**

(E/P) **4** Prove by induction that for all positive integers n, $8^n - 3^n$ is divisible by 5. **(6 marks)**

(E/P) **5** Prove by induction that for all positive integers n, $3^{2n+2} + 8n - 9$ is divisible by 8. **(6 marks)**

(E/P) **6** Prove by induction that for all positive integers n, $2^{6n} + 3^{2n-2}$ is divisible by 5. **(6 marks)**

8.3 Proving statements involving matrices

You can use matrix multiplication to prove results involving powers of matrices.

Example 7

Prove by induction that for all positive integers n, $\begin{pmatrix} 1 & -1 \\ 0 & 2 \end{pmatrix}^n = \begin{pmatrix} 1 & 1 - 2^n \\ 0 & 2^n \end{pmatrix}$.

$n = 1$: LHS $= \begin{pmatrix} 1 & -1 \\ 0 & 2 \end{pmatrix}^1 = \begin{pmatrix} 1 & -1 \\ 0 & 2 \end{pmatrix}$

RHS $= \begin{pmatrix} 1 & 1 - 2^1 \\ 0 & 2^1 \end{pmatrix} = \begin{pmatrix} 1 & -1 \\ 0 & 2 \end{pmatrix}$

As LHS = RHS, the matrix equation is true for $n = 1$.

Assume that the matrix equation is true for $n = k$:

$\begin{pmatrix} 1 & -1 \\ 0 & 2 \end{pmatrix}^k = \begin{pmatrix} 1 & 1 - 2^k \\ 0 & 2^k \end{pmatrix}$

1. Basis step

Substitute $n = 1$ into both the LHS and RHS of the formula to check to see if the formula works for $n = 1$.

2. Assumption step

In this step you assume that the general statement given is true for $n = k$.

With $n = k + 1$ the matrix equation becomes

$$\begin{pmatrix}1 & -1\\ 0 & 2\end{pmatrix}^{k+1} = \begin{pmatrix}1 & -1\\ 0 & 2\end{pmatrix}^{k}\begin{pmatrix}1 & -1\\ 0 & 2\end{pmatrix}$$

$$= \begin{pmatrix}1 & 1-2^k\\ 0 & 2^k\end{pmatrix}\begin{pmatrix}1 & -1\\ 0 & 2\end{pmatrix}$$

$$= \begin{pmatrix}1+0 & -1+2-2(2^k)\\ 0+0 & 0+2(2^k)\end{pmatrix}$$

$$= \begin{pmatrix}1 & 1-2^{k+1}\\ 0 & 2^{k+1}\end{pmatrix}$$

Therefore the matrix equation is true when $n = k + 1$.

If the matrix equation is true for $n = k$, then it is shown to be true for $n = k + 1$. As the matrix equation is true for $n = 1$, it is also true for all $n \in \mathbb{Z}^+$ by mathematical induction.

3. Inductive step

Use the assumption step.

As this is a proof, you should show your working for each element in the matrix multiplication.

← Section 6.2

This is the right-hand side of the original equation with n replaced by $k + 1$.

4. Conclusion step

Example 8

Prove by induction that for all positive integers n, $\begin{pmatrix}-2 & 9\\ -1 & 4\end{pmatrix}^n = \begin{pmatrix}-3n+1 & 9n\\ -n & 3n+1\end{pmatrix}$.

$n = 1$: LHS $= \begin{pmatrix}-2 & 9\\ -1 & 4\end{pmatrix}^1 = \begin{pmatrix}-2 & 9\\ -1 & 4\end{pmatrix}$

RHS $= \begin{pmatrix}-3(1)+1 & 9(1)\\ -(1) & 3(1)+1\end{pmatrix} = \begin{pmatrix}-2 & 9\\ -1 & 4\end{pmatrix}$

As LHS = RHS, the matrix equation is true for $n = 1$.

Assume that the matrix equation is true for $n = k$:

$$\begin{pmatrix}-2 & 9\\ -1 & 4\end{pmatrix}^k = \begin{pmatrix}-3k+1 & 9k\\ -k & 3k+1\end{pmatrix}$$

With $n = k + 1$ the matrix equation becomes

$$\begin{pmatrix}-2 & 9\\ -1 & 4\end{pmatrix}^{k+1} = \begin{pmatrix}-2 & 9\\ -1 & 4\end{pmatrix}^{k}\begin{pmatrix}-2 & 9\\ -1 & 4\end{pmatrix}$$

$$= \begin{pmatrix}-3k+1 & 9k\\ -k & 3k+1\end{pmatrix}\begin{pmatrix}-2 & 9\\ -1 & 4\end{pmatrix}$$

$$= \begin{pmatrix}6k-2-9k & -27k+9+36k\\ 2k-3k-1 & -9k+12k+4\end{pmatrix}$$

$$= \begin{pmatrix}-3k-2 & 9k+9\\ -k-1 & 3k+4\end{pmatrix}$$

$$= \begin{pmatrix}-3(k+1)+1 & 9(k+1)\\ -(k+1) & 3(k+1)+1\end{pmatrix}$$

Therefore the matrix equation is true when $n = k + 1$.

1. Basis step

Substitute $n = 1$ into both the LHS and RHS of the formula to check to see if the formula works for $n = 1$.

2. Assumption step

In this step you assume that the general statement given is true for $n = k$.

3. Inductive step

Use the assumption step.

This is the right-hand side of the original equation with n replaced by $k + 1$.

If the matrix equation is true for $n = k$, then it is shown to be true for $n = k + 1$. As the matrix equation is true for $n = 1$, it is also true for all $n \in \mathbb{Z}^+$ by mathematical induction.

4. Conclusion step

Exercise 8C

E/P **1** Prove by induction that for all positive integers n,

$$\begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}^n = \begin{pmatrix} 1 & 2n \\ 0 & 1 \end{pmatrix}$$

(6 marks)

E/P **2** Prove by induction that for all positive integers n,

$$\begin{pmatrix} 3 & -4 \\ 1 & -1 \end{pmatrix}^n = \begin{pmatrix} 2n+1 & -4n \\ n & -2n+1 \end{pmatrix}$$

(6 marks)

E/P **3** Prove by induction that for all positive integers n,

$$\begin{pmatrix} 2 & 0 \\ 1 & 1 \end{pmatrix}^n = \begin{pmatrix} 2^n & 0 \\ 2^n - 1 & 1 \end{pmatrix}$$

(6 marks)

E/P **4 a** Prove by induction that for all positive integers n,

$$\begin{pmatrix} 5 & -8 \\ 2 & -3 \end{pmatrix}^n = \begin{pmatrix} 4n+1 & -8n \\ 2n & 1-4n \end{pmatrix}$$

(6 marks)

b Hence find the value of n such that:

$$\begin{pmatrix} -1 & 3 \\ -2 & 5 \end{pmatrix}\begin{pmatrix} 5 & -8 \\ 2 & -3 \end{pmatrix}^n = \begin{pmatrix} 11 & -21 \\ 10 & -19 \end{pmatrix}$$

(4 marks)

E/P **5** The matrix $\mathbf{M} = \begin{pmatrix} 2 & 5 \\ 0 & 1 \end{pmatrix}$

a Prove by induction that for all positive integers n,

$$\mathbf{M}^n = \begin{pmatrix} 2^n & 5(2^n - 1) \\ 0 & 1 \end{pmatrix}$$

(6 marks)

b Hence find an expression for $(\mathbf{M}^n)^{-1}$ in terms of n. **(4 marks)**

Challenge

Prove by induction that for all $n \in \mathbb{Z}^+$,

$$\begin{pmatrix} 3 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & -1 & 4 \end{pmatrix}^n = \begin{pmatrix} 3^n & \frac{3^n - 1}{2} & 0 \\ 0 & 1 & 0 \\ 0 & \frac{1 - 4^n}{3} & 4^n \end{pmatrix}$$

Mixed exercise 8

E/P **1** Prove by induction that $9^n - 1$ is divisible by 8 for all positive integers n. **(6 marks)**

P **2** The matrix $\mathbf{B}$ is given by $\mathbf{B} = \begin{pmatrix} 1 & 0 \\ 0 & 3 \end{pmatrix}$.

a Find $\mathbf{B}^2$ and $\mathbf{B}^3$.

b Use your answer to part **a** to suggest a general statement for $\mathbf{B}^n$, for all positive integers n.

c Prove by induction that your answer to part **b** is correct.

E/P **3** Prove by induction that for all positive integers n, $\sum_{r=1}^{n}(3r + 4) = \frac{1}{2}n(3n + 11)$. **(6 marks)**

E/P **4** The matrix $\mathbf{A}$ is given by $\mathbf{A} = \begin{pmatrix} 9 & 16 \\ -4 & -7 \end{pmatrix}$.

a Prove by induction that $\mathbf{A}^n = \begin{pmatrix} 8n + 1 & 16n \\ -4n & 1 - 8n \end{pmatrix}$ for all positive integers n. **(6 marks)**

The matrix $\mathbf{B}$ is given by $\mathbf{B} = (\mathbf{A}^n)^{-1}$.

b Hence find $\mathbf{B}$ in terms of n. **(4 marks)**

E/P **5** The function f is defined by $f(n) = 5^{2n-1} + 1$, where n is a positive integer.

a Show that $f(n + 1) - f(n) = \mu(5^{2n-1})$, where μ is an integer to be determined. **(3 marks)**

b Hence prove by induction that $f(n)$ is divisible by 6. **(4 marks)**

E/P **6** Prove by induction that $7^n + 4^n + 1$ is divisible by 6 for all positive integers n. **(6 marks)**

E/P **7** Prove by induction that for all positive integers n, $\sum_{r=1}^{n} r(r + 4) = \frac{1}{6}n(n + 1)(2n + 13)$. **(6 marks)**

E/P **8** **a** Prove by induction that for all positive integers n:

$$\sum_{r=1}^{2n} r^2 = \frac{1}{3}n(2n + 1)(4n + 1)$$ **(6 marks)**

b Given that $\sum_{r=1}^{2n} r^2 = k\sum_{r=1}^{n} r^2$, show that k must satisfy $n = \dfrac{2 - k}{k - 8}$ **(5 marks)**

E/P **9** The matrix $\mathbf{M} = \begin{pmatrix} 2c & 1 \\ 0 & c \end{pmatrix}$ for some positive constant c

a Prove by induction that for all positive integers n:

$$\mathbf{M}^n = c^n\begin{pmatrix} 2^n & \dfrac{2^n - 1}{c} \\ 0 & 1 \end{pmatrix}$$ **(7 marks)**

b Given that $\det(\mathbf{M}^n) = 50^n$, find the value of c. **(5 marks)**

Challenge

$\mathbf{M} = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$

a Prove by induction that for all positive integers n, $\mathbf{M}^n = \begin{pmatrix} \cos n\theta & -\sin n\theta \\ \sin n\theta & \cos n\theta \end{pmatrix}$.

b Interpret this result geometrically by describing the linear transformations represented by $\mathbf{M}$ and $\mathbf{M}^n$.

Summary of key points

1 You can use **proof by induction** to prove that a general statement is true for all positive integers.

2 Proof by mathematical induction usually consists of the following four steps:

- **Basis:** Show the general statement is true for $n = 1$.
- **Assumption:** Assume that the general statement is true for $n = k$.
- **Inductive:** Show the general statement is true for $n = k + 1$.
- **Conclusion:** State that the general statement is then true for all positive integers, n.